Project Plan

PINN_M3DC1

Physics-informed surrogate architecture for M3D-C1-shaped nonlinear resistive-MHD workflows

Purpose. This is the living roadmap for `PINN_M3DC1`. The project begins with a verified two-dimensional reduced-MHD benchmark and a fair data-only-versus-PINN comparison. Genuine M3D-C1 exports enter only through a later adapter and require a new validation claim.

Current status. Physics notes for Modules 1 through 5 are complete. Module 2 freezes the synthetic solver and verification specification; Module 3 freezes canonical cases, provenance, whole-case splits, preprocessing, versioning, and the M3D-C1 adapter boundary; Modules 4 and 5 freeze the fair data-only-versus-PINN experiment. All numerical and learning modules remain implementation scaffolds: no MHD solver, data pipeline, or trained surrogate is yet claimed.

Contents

1 Objective and claim boundary

The objective is to determine whether physics-informed learning improves a parameterized neural surrogate for nonlinear resistive-MHD evolution under controlled data budgets. Version 1 uses independently generated synthetic arrays. It supports the claim that a PINN can approximate a stated reduced model; it does not establish fidelity to M3D-C1. A validated M3D-C1 surrogate requires genuine code outputs, complete metadata, held-out M3D-C1 cases, and renewed verification.

2 Initial physics benchmark

The periodic two-field model evolves magnetic flux ψ and stream function U. Define

$$[f, g] = f_x g_y - f_y g_x, \qquad J = -\nabla^2\psi, \qquad \omega = \nabla^2 U. \tag{2.1}$$

The frozen version-1 residuals are

$$R_\psi = \partial_t\psi + [U, \psi] - \eta\nabla^2\psi, \tag{2.2}$$

$$R_\omega = \partial_t\omega + [U, \omega] - [J, \psi] - \nu\nabla^2\omega. \tag{2.3}$$

The first nonlinear case is magnetic-island coalescence. It is small enough for independent implementation while retaining advection, current-sheet formation, resistive reconnection, dissipation, and parameter-dependent time evolution.

3 Module roadmap

No.	Module	Purpose
1	Reduced Resistive-MHD Physics and Conventions	Freeze variables, signs, normalization, boundary conditions, invariants, and diagnostic definitions.
2	Synthetic MHD Solver and Case Generation	Generate verified periodic island-coalescence trajectories and convergence evidence.
3	Dataset Contracts and M3D-C1-Shaped I/O	Store fields, parameters, coordinates, provenance, splits, and normalization in versioned schemas.
4	Data-Only Surrogate Baseline	Establish the fair parameterized neural baseline on complete held-out parameter cases.
5	Physics-Informed Surrogate	Add PDE, initial, boundary, and optional global-balance constraints to the same learning task.
6	Validation, Diagnostics, and Benchmarking	Compare field accuracy, physical histories, residuals, generalization, runtime, and ablations.
7	M3D-C1 Export Adapter and Transfer Learning	Map genuine exported fields and metadata into the project contract without silently changing physical meaning.

4 Data and model contract

A synthetic case stores (x, y, t), ψ, U, J, ω, η, ν, A_0, normalization, discretization, boundary conditions, solver provenance, and diagnostic histories. The initial surrogate is

$$(x, y, t, \eta, \nu, A_0) \longmapsto (\widehat{\psi}, \widehat{U}), \tag{4.1}$$

with $\widehat{J}$ and $\widehat{\omega}$ derived by automatic differentiation. Plot images are not training data because they omit the numerical field values and metadata needed to define this map.

5 Controlled comparison

Model	Data loss	Physics losses	Role
Reduced-MHD solver	–	Discretized equations	Synthetic reference
Data-only network	yes	no	Fair neural baseline
Parameterized PINN	yes	PDE/IC/BC	Physics-information test

The two neural models use identical case splits, observations, and comparable capacity. Entire (η, ν, A_0) combinations are excluded from training. Interpolation and extrapolation are reported separately.

6 Verification and validation

The reference solver must pass derivative, bracket, periodicity, divergence, convergence, and energy-behavior checks before producing training data. Surrogate evaluation includes relative L_2 and L_∞ errors for all fields, peak-current and current-sheet errors, magnetic and kinetic energy histories, reconnection metrics, PDE residuals, and periodicity. Runtime reporting includes reference generation, training, inference, and break-even query count.

7 Implementation sequence

1. Freeze equations, normalization, case schema, and diagnostic definitions.
2. Implement the independent pseudo-spectral solver in Python and MATLAB.
3. Produce one converged island-coalescence reference trajectory.
4. Generate a parameter sweep and freeze whole-case data splits.
5. Train the data-only parameterized baseline.
6. Debug a single-case PINN, then train the parameterized PINN.
7. Run ablations, generalization tests, uncertainty checks, and cost analysis.
8. Add and validate an adapter only when genuine M3D-C1 arrays are available.

8 Acceptance milestones

M1 – synthetic solver. One command writes a reproducible HDF5 case and diagnostics; refinement and dissipation checks pass.

M2 – fair learning experiment. Data-only and PINN models share cases, samples, capacity assumptions, and scoring.

M3 – held-out validation. Field and physics metrics are reported on unseen parameter combinations, with interpolation separated from extrapolation.

M4 – M3D-C1 transfer. Exported variables, mesh, normalization, closures, and boundary conditions are mapped explicitly; the full test suite is rerun before an M3D-C1 surrogate claim is made.